

Chapter 3 closed at the point where a specific cognitive gap had been identified. Kripke's semantic account of rigid designation is well-motivated but leaves three connected aspects unspecified at the cognitive level: how rigid tracking is sustained by the speaker, what cognitive event constitutes a baptism, and how the causal chain is psychologically realized. The leading existing implementations — Evans's discriminating conception, Recanati's mental files, Devitt's causal-descriptive hybrid — each address part of the gap but share a representational template that, on the reading developed there, stops short of the primitive tracking capacity the thesis requires.

The present chapter introduces the analytical vocabulary in terms of which the rest of the dissertation closes the gap. The framework's basic primitives — the experiential pattern $\{X\}$, the linguistic pattern $\langle x \rangle$, and the coordination operator $\mathcal{R}$ — are set out in detail in the PRU framework document and are not duplicated here. The chapter takes them as given and develops what they support: a functional-mode distinction between x^o and x^c , a treatment of predication as coordination rather than function-application, the notion of local rigidity, and re-readings of two classical modal puzzles. The next section develops the x^o/x^c distinction, the chapter's central analytical move. The section after that treats $\mathcal{R}$ as a coordination operator that unifies what would otherwise be several distinct predicative and referential relations. The third section introduces local rigidity, the cognitive ground of rigid designation as the thesis here understands it. The fourth and fifth sections apply the resulting apparatus to the de re / de dicto distinction and to the necessary a posteriori. Chapter 5 develops the positive account of baptism and the causal chain in the same vocabulary.

1. The x^o/x^c Functional-Mode Distinction

A word, on the account developed here, has two functional modes. As x^o , it coordinates with a particular instance of an experiential pattern $\{X\}$. As x^c , it operates within the linguistic pattern $\langle x \rangle$ — the network of utterances, frames, and inferential relations in which the word figures. The distinction is functional rather than lexical: the same word-form, “bird,” can play either role, and which role it plays in a given utterance depends on what work the word is being asked to do.

Two examples isolate the modes. In “that bird is injured,” “bird” functions as $bird^o$. It coordinates with one bird in the speaker's perceptual field. The utterance is about that bird; substitute another bird and the truth-value can change. In “birds fly,” the word functions as $bird^c$. It operates within the inferential and generic relations the word maintains in $\langle bird \rangle$ — what is said about birds in the community, how the term combines with predicates, what counts as a counterexample. No particular bird is in view. The utterance is about how the word “bird” connects to “fly” within the linguistic pattern, with whatever generic force the form carries.

The two modes can be told apart by what speakers can and cannot do with them. A child who has acquired $bird^o$ can point at a bird and say “bird”; she can re-identify the same bird across short stretches of time. She cannot yet answer “what is a bird?” except by ostension. A child who has acquired $bird^c$ can answer in words: “a bird is an animal that flies.” The difference is not in the word's reference but in the kind of work the word does — coordinating with a particular versus moving through a network of other words. Both are linguistic capacities. Neither reduces to the other.

This last point bears emphasis because it is where the framework most often gets misread. The object-mode x^o is *still language*. It is not a perception, not a piece of experience. Experience belongs to $E\{X\}$ — the activation, in a given encounter, of the perceptual-motor-affective constellation $\{X\}$. The word $bird^o$ does not *consist of* the experience of the bird; it coordinates with it. The two-side structure — $\langle x \rangle$ on the linguistic side, $\{X\}$ on the experiential side, $\mathcal{R}$ as the coordination between them — keeps x^o from collapsing into either pure linguistic relation or pure perceptual content. A speaker using a word in x^o -mode is doing something linguistic, but doing it in a way that requires coordination with what is encountered.

The rest of the chapter depends on this distinction in specific ways. The treatment of predication in the following section turns on whether the subject of a predication is in x^o or x^c mode. The argument for local rigidity in the section after that locates the cognitive ground of Kripkean rigidity in x^o rather than in x^c . The re-readings of de re/de dicto and of the necessary a posteriori turn on what gets evaluated at each level. The distinction is not introduced as a curiosity; it is the analytical

apparatus the rest of Chapter 4 uses.

The closest neighbor in the existing literature is Recanati's mental-files architecture. Recanati holds that singular thought is carried by mental files — vehicles of reference in a language-of-thought architecture, typed by the kind of relation that grounds them.¹ A demonstrative file is opened in a perceptual encounter and stores information from that encounter; a memory-demonstrative file is what the demonstrative file becomes once perceptual contact ends; a recognitional file collects information about an individual recognized across encounters; an "encyclopedia entry" is a conceptual file, grounded in a higher-order relation rather than in direct acquaintance, but still — on Recanati's view — based on an *epistemically rewarding* (ER) relation to its referent.² The demonstrative file is the basic case; the others are derivative. The architecture handles a great deal of what the account here also wants to handle: the difference between a thought tied to a present particular and a thought about an absent or recognized one, the way information accumulates around a referent, the conditions under which two singular thoughts converge on a single object.

Three differences between Recanati's distinction and the one drawn here are worth stating clearly. They are not arguments against Recanati; they mark the points at which the account here does different work.

The first concerns the unit of analysis. For Recanati, the unit is the file — a mental representation, "the mental counterpart" of a singular term, operating in a language-of-thought architecture.³ On the account here, the unit is the word, used in one or the other functional mode. The framing is not that the mind has a file *Aristotle* with an ER-grounded slot, but that the public word "Aristotle" functions in two modes which a speaker exercises in conversation, instruction, baptism, dispute. The dissertation is about how the public language refers, not about which cognitive vehicle carries the speaker's singular thought. The two questions can be asked independently, and giving an answer to one does not commit to a particular answer to the other.

The second concerns the topology of the interface. Recanati's files are mental items related to objects in the world through ER relations; the interface is mind-to-world. PRU has a different topology: $\langle x \rangle$ on one side, $\{X\}$ on the other, and $\mathcal{R}$ as the coordination between them. Both x^o and x^c are word-modes living on the linguistic side, but x^o is the mode in which the word coordinates, through $\mathcal{R}$, with a particular activation $E\{X\}$ on the experiential side. The linguistic pattern $\langle x \rangle$ — the publicly maintained pattern of word-use in the community — has no clear counterpart in the file architecture; Recanati's files are private mental items, even when they happen to converge across speakers. The PRU topology takes the social-linguistic dimension of x^c as primitive rather than emergent, which matters when Chapter 5 treats baptism and the causal chain as operations on $\langle N \rangle$ rather than as transfers of files.

The third concerns the treatment of common nouns. Recanati's taxonomy is built for singular reference. Encyclopedia entries are normatively "one per object of interest"; demonstrative and memory files are tied to particular objects through ER relations. Common kind-terms — "bird," "water," "chair" — do not fit the file architecture without strain, and the difficulty has been noticed from within the file-theoretic literature.⁴ The account here applies the x^o/x^c distinction symmetrically to common nouns and to names: a kind-term used to coordinate with a particular ("that bird") is in object-mode; the same kind-term used in inferential and generic relations ("birds fly") is in concept-mode. Names, treated in Chapter 5, emerge through a baptismal lifting of h^o into a stabilized N^c , but the underlying machinery — the x^o/x^c functional distinction — is the same one already at work for common nouns. The symmetry matters for the argument that names scale up demonstrative rigidity, because that argument needs demonstrative coordination already to be present at the level of common nouns rather than emerging only with the use of proper names.

What the distinction earns. The originality of the move is not in noticing that some uses are tied to particulars while others are not; that observation is at least as old as Russell's distinction between knowledge by acquaintance and knowledge by description and runs through Strawson, Evans, and Recanati. The move is in locating the distinction at the level of word-modes, with

¹Recanati, *Mental Files* (2012), ch. 3.

²Recanati, *Mental Files* (2012), chs. 3–4 on the typology of files; ch. 7 on encyclopedia entries. The view that even encyclopedia entries are grounded in ER relations is the "actualist" commitment criticized by Hansen and Rey in F. Salis (ed.), *Disputatio V* (36), 2013.

³Recanati, *Mental Files* (2012), p. 34: "A non-descriptive mode of presentation, I claim, is nothing but a mental-file." The vehicle/content framing is explicit at pp. 31–34.

⁴Goodman, "Against the Mental Files Conception of Singular Thought," *Review of Philosophy and Psychology* 7 (2016): 437–61, presses a related point from a different angle: that some file-based thought is descriptive rather than singular, so the identification of file-based thought with singular thought breaks down.

$\mathcal{R}$ as the coordination between the linguistic side and the experiential side, and with the same machinery applying to common nouns and to names. This locates the cognitive ground of rigid designation in something already at work in ordinary demonstrative speech, without requiring a separate vehicle-theoretic apparatus and without restricting the analysis to singular thought narrowly construed. The remaining sections of Chapter 4 put the distinction to work; Chapter 5 uses it to give the positive account of baptism, the causal chain, and the metasemantic role of $\mathcal{R}$.

2. $\mathcal{R}$ as Coordination

The relation $\mathcal{R}$ is the framework’s coordination operator. It describes how the items of the two-side topology — words in $\langle x \rangle$ and patterns in $\{X\}$ — stand in relation to one another. The relation can be of several kinds, depending on what is being coordinated. A word in object-mode can coordinate with an experiential activation: $\mathcal{R}(\text{bird}^o, E\{BIRD\})$ describes the binding by which “bird” used demonstratively coordinates with a particular bird in view. A word in object-mode can coordinate with a word in concept-mode: $\mathcal{R}(\text{bird}^o, \text{white}^c)$ describes the predicative coordination by which “that bird is white” attaches the concept-mode predicate to the object-mode subject. A word in concept-mode can coordinate with another word in concept-mode: $\mathcal{R}(\text{bird}^c, \text{fly}^c)$ describes the generic coordination by which “birds fly” connects the two concept-mode items within the linguistic pattern.⁵

In each case the operator names the same kind of phenomenon — two items standing in coordination — and the argument types determine the specific kind. The claim of the framework is that what looks like four different things in standard semantics — reference, predication of a particular, generic predication, and (in Chapter 5) baptism — are four applications of one relation. Whether this is borne out depends on whether the unified treatment handles the cases naturally and produces the right consequences for the dissertation’s downstream arguments. The remaining sections of Chapter 4 and the whole of Chapter 5 are the test.

The word “coordination” carries the substantive load here. The standard alternative is Fregean function-application: a predicate is a function that takes an object as argument and yields a truth-value. Set-theoretic semantics produces a parallel architecture: a predicate is a set, predication is membership. Both pictures are asymmetric and arity-fixed: the predicate is the active item, the object the passive one, and the predication relation is the application of one to the other. The coordination picture differs at exactly this point. The two items in $\mathcal{R}(x, y)$ stand in a structural relation that the practice sustains, not in a function-argument pairing the speaker computes. The asymmetry between “subject” and “predicate” in surface grammar is real but not constitutive: in $\mathcal{R}(\text{bird}^c, \text{fly}^c)$ neither item is more “argument” than the other, and the same operator covers the case where one item is in object-mode and the other in concept-mode. The unification is what the coordination framing buys.

This is not a computational picture. $\mathcal{R}$ is not an operation a speaker *performs* on linguistic items, in the way a function gets applied to an argument or a set-membership relation gets checked. It is a relation that holds in virtue of how the speaker is embedded in a practice. To say that $\mathcal{R}(\text{bird}^o, \text{white}^c)$ holds is to say that the speaker’s use of “white” of this bird is sustained by ongoing engagement with both the bird (perceptually) and the predicate (within $\langle \text{white} \rangle$, where the term has its application conditions, inferential roles, and contrast cases). The coordination is constituted developmentally — through the speaker’s acquisition of the relevant patterns — and is sustained by continued participation in the linguistic community and continued contact with the world. The register is closer to Wittgenstein than to cognitive engineering: $\mathcal{R}$ describes a feature of how language is lived, not a procedure executed in real time.

The closest neighbors in the existing literature are Sellars’s account of language-entry and language-exit transitions, together with Brandom’s inferentialism. Sellars proposed that the use of a word is governed by three kinds of normative connection: language-entry rules (linking words to perceptual circumstances), language-language rules (linking words to other words through material inference), and language-exit rules (linking words to action).⁶ Brandom developed the language-language side into a thoroughgoing inferentialism: the content of a concept is constituted by its inferential role in the network of material commitments and entitlements that a competent speaker can give and ask

⁵ $\mathcal{R}$ is specified in the framework document at greater technical length, including its developmental account in the learning of $\{X\}$ and its role in the integrated $\{X, x^o\}$ constellation produced by the coordination of experiential event and word; see PRU Meta-Language, version 6.

⁶Sellars, “Some Reflections on Language Games,” *Philosophy of Science* 21 (1954): 204–28; reprinted in *Science, Perception and Reality* (1963), pp. 321–58. The three-fold scheme of entry, intra-linguistic, and exit transitions is developed in §§II–III of that essay.

for.⁷ On the resulting picture, the use of “red” of an apple is jointly governed by language-entry rules (you can call it “red” because of how it looks) and inferential roles (calling something “red” commits you to its not being green, to its being colored, and so on).

The relation between this picture and $\mathcal{R}$ is close enough to require comment, and the points of departure are specific. Sellars and Brandom maintain a two-part architecture: entry and exit transitions on one side, inferential roles on the other. The world-language coupling and the word-word inferential coupling are distinct kinds of normative connection, governed by different sorts of rules. The proposal here unifies them. The same operator $\mathcal{R}$ handles $\mathcal{R}(bird^o, E\{BIRD\})$ (the language-entry side, in Sellars’s terms) and $\mathcal{R}(bird^c, fly^c)$ (the inferential side). The unification claim is that these are one kind of relation applied to different argument types, not two kinds of relation that happen to operate at the same level. Whether one finds this convincing depends on whether one finds the cases genuinely structurally similar; the dissertation’s argument is that they are, and that the rigid-designation phenomena worked out in the next section and in Chapter 5 are best handled by an apparatus that treats them as such.

Two further points of departure are worth flagging. Brandom’s inferentialism takes inferential role as constitutive of conceptual content; $\mathcal{R}$ is neutral on this question. The coordination relation can hold whether content is constituted by inference, by truth-conditions, by use, or by some other structure. The framework here does not need to settle the Brandom-versus-truth-conditional debate to do its work. Second, Sellars and Brandom both give privileged status to assertion as the unit of inferential commitment. $\mathcal{R}$ takes word-modes as its arguments, and these are smaller-than-assertion units. The chapter’s downstream applications — to rigid designation, de re/de dicto, the necessary a posteriori — turn on relations between sub-assertoric word-uses rather than between assertions, which is one reason the assertion-centric framings do not quite fit the work the dissertation needs done.

With $\mathcal{R}$ in place, the chapter can handle predication, reference, and conceptual coordination with one piece of machinery, and Chapter 5 can extend the same machinery to baptism and the causal chain without new primitives. The next section uses the apparatus to argue that the cognitive ground of rigid designation appears already at the level of $\mathcal{R}(x^o, \cdot)$ — coordination of an object-mode word with anything else — and that the rigidity of names is the same coordination stabilized for use across absence. The sections after that re-read de re/de dicto and the necessary a posteriori in the same terms.

3. Local Rigidity at x^o

Consider the utterance “that bird could have been larger.” A speaker pointing at a bird in front of her makes the counterfactual claim. The bird in question is held fixed across the modal variation: the speaker is not imagining a different bird that is larger; she is imagining *that very bird* having been larger. The word “bird” in x^o mode coordinates with the same particular across the modal scope of “could have been.” Whatever properties of the bird are varied — size, color, vigor, age — the bird stays the same, by virtue of the demonstrative coordination already in place.

The modal test for rigid designation is whether a designator picks out the same individual across the worlds the modal context quantifies over.⁸ The test, applied here, passes. In every counterfactual scenario where “that bird could have been larger” is evaluated, “that bird” picks out the same bird that is in front of the speaker in the actual context. The variation occurs in what is predicated of the bird, not in which bird is being talked about. This is what rigid designation looks like at the level of demonstrative coordination.

Call this *local rigidity*: rigidity that obtains within a perceptual or conversational episode by virtue of the demonstrative coordination $\mathcal{R}(x^o, \cdot)$ being in place. Local rigidity is not yet Kripkean rigidity proper. Kripkean rigidity attaches to a name and holds across the entire range of counterfactual scenarios in which the bearer exists; local rigidity holds within the episode and lapses when the coordination does — when the speaker stops attending to the bird, or when the conversational context shifts. The two differ in scope, not in structural kind. What is varied is the durability of the coordination, not the cognitive operation that constitutes the coordination.

This last point bears emphasis, because the structural claim of the chapter depends on it. The claim is that rigidity is not a stiff property of the designator. It is a preserved link between the

⁷Brandom, *Making It Explicit* (1994), esp. chs. 3–4; *Articulating Reasons* (2000), ch. 1. For critique of Brandom’s account of singular terms specifically — relevant since singular reference is the topic of the present chapter — see McCullagh, “Inferentialism and Singular Reference,” *Canadian Journal of Philosophy* 35 (2005): 183–220.

⁸Kripke, *Naming and Necessity*, p. 48. The test was developed in Chapter 3 §3.1.

speaker's use of the term and the particular that anchors the use. The preservation is constituted by the practice — perceptual attention, in the demonstrative case; social-linguistic stabilization, in the case of names. To call a term rigid is to say that this link is being preserved across whatever variation the modal context introduces. The variation is in the properties imagined, not in the link itself.

It is worth setting this picture against the natural alternative. A different way of conceiving rigidity treats the designator as something with rigidity built in — a kind of stiff attachment to its bearer that, once established, holds across modal variation by virtue of what the designator is. On that picture, what makes the modal evaluation come out right is the rigidity of the attachment itself, a feature the designator possesses. The picture is a natural one and is encouraged by some of Kripke's own formulations, but it misdescribes what is happening. Names do not have rigidity built in. Names rigidly designate because the linguistic community has a practice of using them rigidly — of correcting failures of reference, of holding the term to its bearer across descriptive revision, of teaching the term so that successive speakers continue to use it in the same way.⁹ The preserved-link framing captures what this practice accomplishes; the built-in-property framing obscures it.

The consequence for what names are. Names, on the account developed here, are devices for sustaining the preserved link across the conditions that make demonstrative re-anchoring impossible: the bearer is not perceptually present, the speaker is not the baptizer, the chain of transmission is many speakers and centuries long. The cognitive coordination that operates in $\mathcal{R}(\text{bird}^o, \cdot)$ at the level of demonstrative use is the same coordination that operates in $\mathcal{R}(\sigma, \text{Aristotle}^c)$ at the level of a surrogate-based reading of Aristotle two millennia after his death.¹⁰ What differs is the scaffolding that sustains the coordination: perceptual contact in the first case, testimonial networks and surrogate objects (manuscripts, portraits, citations) in the second. The cognitive operation is the same; the practice that preserves the link is different. Chapter 5 develops this in detail, treating baptism as the operation that lifts the local coordination $\mathcal{R}(h^o, N^c)$ into a socially portable form and the causal chain as the testimonial infrastructure that maintains the lift.

The closest neighbor in the existing literature is the indexicalist tradition on proper names. Russell's late view in *Inquiries into Meaning and Truth* already treated names as a class of indexicals; Recanati's *Direct Reference* developed the picture systematically; Pelczar and Rainsbury treated names as pure indexicals with a Kaplanian "character" assigning dubbings to contexts; Rami developed a use-conditional indexical conception; Kölne and Textor proposed names as hybrid expressions composed of an utterance and its referent; Tarnowski argues that proper names are simple demonstratives whose referents are determined by speaker's referential intention.¹¹ On these views, names belong to the semantic category of indexicals or demonstratives — they have a character function from contexts to contents, and their reference is determined by features of the context of use rather than by descriptions associated with the term.

The position taken here is not a position in this debate. The indexicalist tradition is doing *semantic* work: it asks what category names belong to and how their reference is determined by context. Its central explanatory target is *nam ambiguity* — the phenomenon of multiple bearers of the same name — and its central technical move is to give names a Kaplan-style character. The local-rigidity claim is a *metasemantic* claim about the cognitive coordination that supports rigid designation, whatever semantic theory of names one favours. The PRU view is compatible with Kaplanian semantics for names, with Kripke's causal-chain picture, with Millian direct reference. What it adds is a story about the cognitive operation any of these accounts presupposes when it speaks of a name "tracking" a bearer across worlds, contexts, or speakers.

Three points of departure from the indexicalist tradition are worth marking. The first concerns

⁹Kripke's own metaphors are not in fact uniformly committed to a built-in-property reading. His description of the baptismal act and the causal chain (NN pp. 91–97) treats reference as something speakers do and sustain, which is closer to the preserved-link framing. The built-in-property reading is a tendency of the formal regimentation rather than of Kripke's positive picture; see Kripke's own characterization of his account as "a better picture" rather than a theory at NN p. 94, treated in Chapter 3 §3.4.

¹⁰The notation σ for surrogate objects — manuscripts, portraits, citations — is from the PRU framework document. The surrogate-based reading of names without direct experiential contact is developed in Chapter 5.

¹¹Russell, *Inquiries into Meaning and Truth* (1940); Recanati, *Direct Reference: From Language to Thought* (1993); Pelczar and Rainsbury, "The Indexical Character of Names," *Synthese* 114 (1998): 293–317; Rami, "The Use-Conditional Indexical Conception of Proper Names," *Philosophical Studies* 168 (2014): 119–50; Kölne, "Hybrid Proper Names," *Mind* 101 (1992): 721–31; Textor, "Frege's Theory of Hybrid Proper Names Extended," *Mind* 124 (2015): 823–47; Tarnowski, "Proper Names as Demonstratives," in C. Penco and A. Negro (eds.), *Proceedings of the 2021 Workshop on Context*. For a survey of the broader field, see García-Carpintero, "The Mill-Frege Theory of Proper Names," *Mind* 127 (2018): 1107–68.

the level of analysis: the indexicalist tradition gives names a semantic category and revises Kaplan-style semantics accordingly; the present account does not revise Kripkean or Kaplanian semantics at all. It addresses a question those semantics presuppose without answering. The second concerns the explanatory target: the indexicalist tradition's central case is nambiguity and the same-name-multiple-bearers phenomenon; the present account's central case is the cognitive ground of Kripkean rigidity, which is a different problem.¹² The third concerns common nouns: the indexicalist accounts are tailored to proper names specifically and do not naturally generalize to "that bird," "this dog," or other common kind-terms used demonstratively. The present account treats the demonstrative use of "bird" and the use of "Aristotle" as cases of the same underlying coordination, differing in the practice that sustains the coordination rather than in the kind of cognitive operation involved.

With local rigidity in place and the preserved-link framing developed, the chapter has a unified account of what rigidity is at the cognitive level. The same coordination operates in demonstrative use and in name-use; what differs is the social-linguistic infrastructure that sustains the coordination across absence. The next section uses the level distinction to re-read *de re* and *de dicto* modality. The section after that handles the necessary *a posteriori* as a case where the same individual is held fixed at x^o (necessary) while the inference connecting two concept-mode designators of that individual requires empirical work (*a posteriori*). Chapter 5 develops the positive account of baptism and the causal chain as the practice that lifts local rigidity into the durable form proper names exhibit.

4. De Re and De Dicto as Predication at x^o and x^c

The *de re* / *de dicto* distinction is medieval in origin and gets its modern form in Quine's puzzle about quantifying into modal contexts.¹³ Consider the canonical example:

Necessarily, the number of planets is greater than seven.

The sentence is ambiguous. On one reading — *de dicto*, or narrow scope — the description "the number of planets" is evaluated at each world. In every world, the number of planets in that world must be greater than seven. The reading is false: there are possible worlds with fewer than seven planets. On the other reading — *de re*, or wide scope — the description picks out the actual number of planets, eight, and the modal claim is about that number. Of eight, necessarily it is greater than seven. The reading is true.

Two standard treatments handle this. Russell's scope-of-the-quantifier solution treats the ambiguity as one of logical form: the description can take scope inside or outside the modal operator, and the two scopes yield the two readings.¹⁴ Kripke's rigidity solution treats it as a feature of the difference between names and descriptions: names are rigid and behave like wide-scope descriptions in modal contexts; descriptions can be evaluated either way.¹⁵ The two treatments are not in conflict; they handle the same phenomenon at different levels of theory.

The present reading takes the ambiguity to be a manifestation of the x^o/x^c mode distinction. On this reading, "the number of planets" can function in either mode. In x^c mode, the description operates within $\langle \text{number of planets} \rangle$ — the linguistic pattern in which the term has its conditions of application, evaluated world by world. This is the *de dicto* reading. In x^o mode, the description coordinates with the particular it picks out in the actual context — the number eight — and that particular is held fixed across the modal evaluation. This is the *de re* reading. The ambiguity is not, on this view, a feature of the modal operator's scope or of a category difference between names and descriptions; it is a feature of which mode the singular term is being used in.

A few consequences are worth noting. The first: the *de re* / *de dicto* distinction is not specific to descriptions. Any term that can be used in either mode admits the distinction. "Birds, necessarily, are not amphibians" has a *de dicto* reading (whatever the practice picks out as "birds" in each world is, by the practice's standards, not amphibian) and a *de re* reading (of birds — the actual

¹²This is not to say the present account has nothing to say about nambiguity. Multiple-bearer cases are handled by the linguistic pattern $\langle N \rangle$ admitting multiple anchorings $\mathcal{R}(h_1^o, N^c)$, $\mathcal{R}(h_2^o, N^c)$, etc., with context determining which anchoring is operative in a given use. The question is downstream of the local-rigidity claim, and is not the present section's main concern.

¹³Quine, "Reference and Modality," in *From a Logical Point of View* (1953); Kaplan, "Quantifying In," *Synthese* 19 (1968): 178–214. The medieval background is reviewed in Knuuttila, *Modalities in Medieval Philosophy* (1993).

¹⁴Russell, "On Denoting," *Mind* 14 (1905): 479–93, on scope generally; the modal application is developed in Smullyan, "Modality and Description," *Journal of Symbolic Logic* 13 (1948): 31–37.

¹⁵Kripke, *Naming and Necessity*, pp. 48–60 and the scope footnote at p. 12. Chapter 3 §3.2 worked through the scope material in detail.

ones, which form a particular natural kind — necessarily, none of them is an amphibian). The reading depends on whether “birds” is operating in x^c mode within the linguistic pattern ⟨birds⟩ or in x^o mode, coordinating with the actual kind.¹⁶

The second: rigid designation, on this reading, is the case where a term is *exclusively* in x^o mode for modal evaluation. Proper names, as Chapter 3 argued, function this way: “Aristotle” has no de dicto reading, no reading on which the modal evaluation re-picks-out whoever fills certain conditions in each world. The term is in x^o mode constitutively, by virtue of the baptismal coordination that fixed it. Rigidity, on this picture, is what happens to the modal behavior of a term when its mode is invariant across modal evaluation.

The third: Quine’s worry about quantifying into modal contexts was that essentialism — the idea that things have essential properties independent of how they are described — was metaphysically suspect.¹⁷ The present reading addresses this worry indirectly. The de re reading does not require essentialism in Quine’s sense. It requires only that the speaker is using the term in x^o mode, coordinating with a particular and holding the coordination fixed across modal variation. What properties the particular has across worlds is a separate question, on which the framework here takes no stand. Quine’s worry assumed that de re modality presupposes a metaphysics of essential properties; the present account locates the de re reading in the cognitive operation of holding a particular fixed, not in any metaphysical commitment about what the particular essentially is.

The unification this section achieves is modest. The de re / de dicto distinction has been handled before, in scope-theoretic terms and in rigidity-theoretic terms. What is added by the present reading is that the distinction is one consequence of the wider x^o/x^c mode distinction — the same level distinction that handles rigid designation in the previous section and the necessary a posteriori in the next. The de re / de dicto distinction is not, on this view, a separate problem with its own solution; it is what the level distinction looks like when the modal operator is applied to a non-name singular term.

5. The Necessary A Posteriori at the Level Distinction

Kripke’s claim that some identities are necessary but a posteriori broke a connection that had been taken for granted in much of the analytic tradition: that necessity goes with analyticity and a priori knowability, while a posteriori truths are contingent. The classical examples are identities involving rigid designators — “Hesperus is Phosphorus,” “water is H_2O ,” “this very table is made of wood.” Each is necessary, on Kripke’s account, because both flanking terms rigidly designate the same particular or natural kind. Each is a posteriori because the identity was established empirically.¹⁸

The phenomenon has been read by some as paradoxical. If “Hesperus is Phosphorus” is necessary, why was the identity not known a priori, like “ $2 + 2 = 4$ ”? If it is a posteriori, how can it be necessary rather than contingent? The puzzlement is real on a picture where necessity is identified with analyticity and epistemic accessibility tracks metaphysical modality. On the picture developed in this chapter, the puzzlement has a clean source: it conflates two levels of language use that should be kept apart.

The necessity lives at x^o . When Hesperus is observed in the evening and Phosphorus in the morning, the same celestial body is the object of two demonstrative coordinations — once at dusk, once at dawn. At the level of x^o , there is only one Venus: $E\{VENUS\}$ activated in the evening is, as it happens, $E\{VENUS\}$ activated in the morning. The identity of the planet is constitutive at the demonstrative-coordination level: the same particular is being tracked in both cases. Once it is granted that there is one star out there that the two demonstrative coordinations track, the identity at x^o is immediate. No discovery is needed at this level.

The a posteriority lives at x^c . The concept-mode designators ⟨Hesperus⟩ and ⟨Phosphorus⟩ are distinct linguistic patterns. Each was established by an independent baptismal lifting: one anchored in the evening-sky coordination, the other in the morning-sky coordination. As concept-mode terms they have different usage histories, different inferential connections, different roles

¹⁶Wiggins’s *Sameness and Substance* (1980) develops a related though differently framed point about sortals and individuation. Fine’s “Essence and Modality,” *Philosophical Perspectives* 8 (1994): 1–16, presses a different question: whether essence (rather than modality) is the fundamental notion. Both questions are downstream of the level distinction at issue here; the present section makes no commitment on either.

¹⁷Quine, “Reference and Modality” (1953); “Three Grades of Modal Involvement,” *Proceedings of the XIth International Congress of Philosophy* 14 (1953): 65–81.

¹⁸Kripke, *Naming and Necessity*, pp. 100–105 on Hesperus/Phosphorus; pp. 116–18 on water/ H_2O ; the general thesis is at pp. 35–39. The pre-Kripkean connection between necessity, the a priori, and analyticity was defended in different forms by Carnap, Ayer, and others, and is the standard target of Kripke’s first lecture.

in their respective linguistic patterns. Establishing that they coordinate to the same particular requires empirical work — astronomical observation, theoretical inference, and acceptance of the identification by the community. The inference $\mathcal{R}(Hesperus^c, Phosphorus^c)$, that the two concept-mode patterns share a referent, is a posteriori in the standard sense.

The phenomenon is not a paradox on this reading. It is a structural fact about how language operates at two levels. The identity is necessary at the level where the same particular is being coordinated with; the discovery of the identity is a posteriori at the level where the two linguistic patterns are reconciled. There is no single statement that is both necessary and a posteriori in a single sense; there is a statement that picks up necessity from its x^o aspect and a posteriority from its x^c aspect, and these aspects belong to different levels of language use. The traditional connection between necessity and a priori knowability assumed that there is only one level at which the statement operates. The x^o/x^c distinction blocks the assumption.

The same applies to “water is H_2O .” At the level of x^o , the natural kind whose instances were demonstratively tracked by speakers using “water” is the same natural kind whose instances are chemically composed of H_2O molecules. There is one kind, and the demonstrative tracking by ordinary speakers and the chemical analysis by scientists coordinate to it. The identity is necessary at this level. At the level of x^c , the two terms — the ordinary-language $\langle water \rangle$ and the chemical-formula $\langle H_2O \rangle$ — are different linguistic patterns with different inferential roles. Establishing that they coordinate to the same kind required scientific investigation. The identity is a posteriori at this level.¹⁹

What the section adds. The necessary a posteriori has been handled in the literature through theories of trans-world identity, essentialism about substances and natural kinds, and two-dimensional semantics.²⁰ The present treatment does not contest any of these accounts at the semantic level. It locates the apparent paradox at the cognitive-linguistic level: necessity at x^o , a posteriority at x^c . The two levels were conflated in the tradition Kripke broke from, and they remain easy to conflate even after Kripke. The level distinction marks them apart and gives a structural account of why the necessary-a-posteriori cases have the shape they have.

The chapter closes here. The x^o/x^c distinction has been introduced and motivated; the coordination operator $\mathcal{R}$ has been specified and contrasted with function-application and inferentialist alternatives; local rigidity has been developed and positioned against the indexicalist tradition; the de re / de dicto distinction has been re-read as the modal expression of the level distinction; and the necessary a posteriori has been reconstructed as a structural consequence of the level distinction. The next chapter develops the positive account of baptism, the causal chain, and the metasemantic role of PRU: how speakers come to inhabit Kripkean models, what baptism does at the cognitive level, how surrogate objects and testimonial networks sustain the coordination across the centuries, and why this account complements rather than competes with the semantic apparatus Kripke and his successors developed.

¹⁹The water- H_2O case raises additional questions about natural kinds, essence, and the role of scientific theorizing in fixing reference. These are discussed in Putnam, “The Meaning of ‘Meaning’,” in *Mind, Language and Reality* (1975), and have generated a substantial literature. The present section makes no commitment on the metaphysics of natural kinds; the level distinction handles the modal-epistemic shape of the case without taking a position on whether being H_2O is the essence of water in any deeper sense.

²⁰Salmon, *Reference and Essence* (1981); Chalmers, “The Foundations of Two-Dimensional Semantics,” in *Two-Dimensional Semantics* (2006); and the survey in Soames, *Beyond Rigidity* (2002), chs. 7–9.